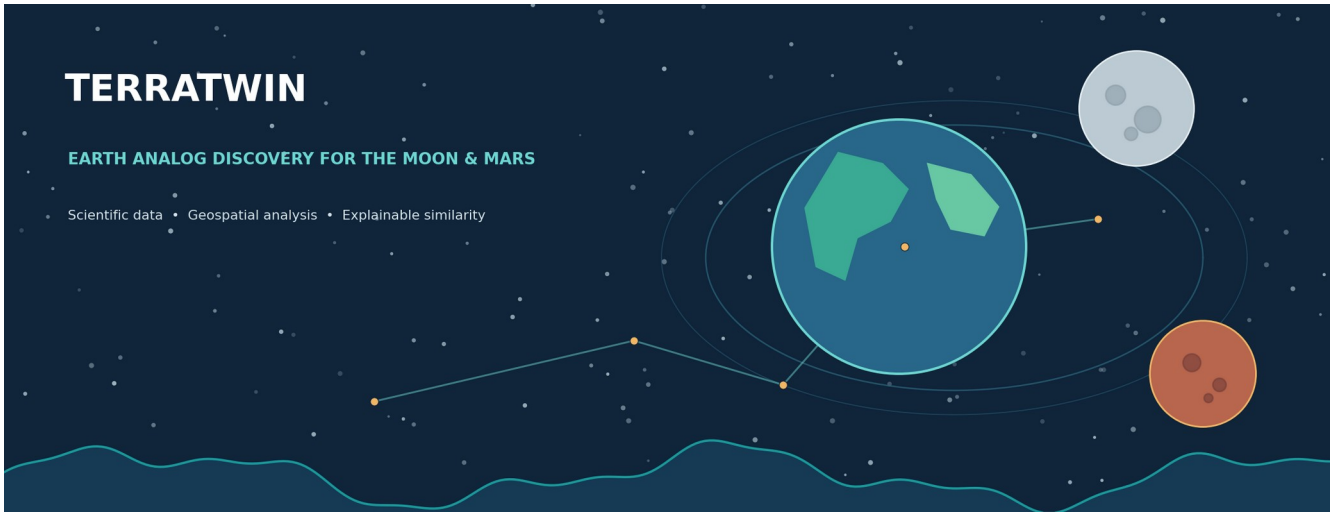


PROJECT REPORT

TerraTwin

Identifying Earth Locations Analogous to Permanent Moon Base Sites and Mars

THE IDEA IN ONE SENTENCE

A geospatial decision-support platform that compares measured Earth environments with selected lunar and Martian reference regions, ranks candidate analogs transparently, and explains where similarities and differences occur.

DOMAIN	PRIMARY OUTPUT	STATUS
Space science + geospatial AI	**Ranked Earth analog candidates**	**Proposed hackathon MVP**

Prepared as a concept and implementation plan for the NASA Space Apps Challenge 2026.

Important: all rankings, scores and candidate findings described in this report are proposed outputs. No live analysis results are claimed.

1. Executive Summary

TerraTwin is a proposed geospatial analytics platform for finding and characterizing Earth locations that share selected, measurable characteristics with lunar and Martian environments. The NASA Space Apps Challenge describes a need to identify and better characterize terrestrial analogs using open Earth, Moon and Mars data. Existing analog environments include deserts, polar regions, volcanic and arid terrain, caves and other landscapes that may resemble particular aspects of planetary surfaces [1].

The system will combine suitable open datasets, Python-based geospatial processing, a transparent similarity-scoring method and an interactive web dashboard. Users will select a planetary reference region, choose comparison features, review ranked Earth candidates and inspect explanations for each result.

CORE VALUE

TerraTwin turns planetary and Earth observation data into a repeatable, explainable workflow for exploring candidate analog environments - instead of merely plotting already-known sites.

1.1 Project at a glance

Item	Description
Project name	TerraTwin
Challenge focus	Identify Earth locations analogous to permanent Moon base locations and Mars
Primary disciplines	Data science, GIS, planetary science, remote sensing, software development
Target users	Planetary researchers, analog-site planners, robotics teams, educators and students
Core input	Open gridded or vector data for Earth and selected lunar/Martian reference regions
Core output	Ranked candidates, feature comparisons, map visualizations, data provenance and uncertainty notes
MVP principle	Small, reproducible and scientifically defensible; no fabricated observations

1.2 Executive objectives

- Integrate a manageable set of compatible Earth, Moon and Mars datasets.
- Compute comparable terrain and environmental features with documented units and resolution.
- Rank Earth candidate regions with a clear, configurable distance method.
- Expose the evidence behind every score, including data gaps and limitations.
- Deliver a demonstrable web app and a reproducible analysis pipeline.

2. Problem, Context and Proposed Solution

2.1 Problem statement

Testing every technology and operational procedure directly on the Moon or Mars is costly and difficult. Earth analog sites can support selected field studies and equipment trials, but an Earth site that looks similar in photographs may differ greatly in slope, roughness, geology, illumination or other relevant factors. The challenge is therefore not only to find visually similar landscapes, but to compare documented measurements and characterize where the analog is useful and where it is not [1].

2.2 Proposed solution

TerraTwin will provide a map-based exploration and comparison workflow. A user selects a lunar or Martian reference area and a set of features appropriate to the research question. The system extracts or loads comparable feature summaries, normalizes differences, ranks candidate Earth regions and presents an evidence-based explanation of each result.

2.3 Typical user workflow

Step	Action	What the system does
01	Choose a reference	Pick a lunar or Martian site and define the region of interest.
02	Choose features	Select variables such as slope, roughness, relative relief, geology or illumination where supported.
03	Process data	Standardize units, coordinate systems, spatial scale and quality flags.
04	Rank candidates	Calculate feature distances and produce a transparent similarity ranking.
05	Inspect evidence	Open maps, charts, match explanations, missing-data notes and source links.

2.4 Intended users and value

User	Potential value
Planetary scientists	Discover and characterize additional candidate analog environments for further research.
Mission and robotics teams	Explore terrain types relevant to mobility, navigation or selected field tests.
Educators and students	Understand planetary environments using real observations and transparent comparisons.
Science communicators	Present complex data as accessible maps, charts and evidence-backed summaries.

SCOPE BOUNDARY

A similarity score is not a safety rating, base-site approval or prediction of astronaut survival. TerraTwin is a decision-support prototype, not a replacement for domain-expert review.

3. Scientific Methodology

3.1 Reference sites

For a manageable first release, use one well-documented lunar region and one Martian region. Shackleton Crater / the lunar south-pole region is a possible lunar reference because topography and illumination are relevant to operations there. Jezero Crater is a possible Mars reference for terrain and geological-context exploration. These are candidate study areas for the project, not claims that they are the best or officially selected sites for a permanent base. Confirm the final region and the available data products before implementation.

3.2 Candidate Earth environments

Start with a limited candidate set drawn from distinct environment types - for example, arid deserts, polar environments, volcanic landscapes, crater-like landforms and exposed rocky terrain. Known analog sites such as the Atacama Desert, Haughton Crater and Death Valley can serve as contextual examples or sanity checks; the project should aim to characterize candidates through data rather than hard-code those sites as winners [1].

3.3 Feature selection

Feature	Possible measure	Interpretation / caveat
Slope	Degrees or distribution of slope	Terrain steepness; derived from compatible elevation products.
Surface roughness	Documented roughness statistic	Local surface complexity; metric and scale must be specified.
Relative relief	Elevation range within a fixed window	Compare local terrain variation, not raw elevation datums.
Landform shape	Crater, valley or ridge morphometry	Useful only where a reproducible feature extraction is available.
Reflectance / albedo	Product-specific values	Compare only after accounting for instrument and calibration differences.
Illumination	Sunlit fraction or modeled shadow duration	Requires appropriate body geometry and time/season assumptions.
Temperature	Range or distribution for defined time windows	Earth weather and planetary surface temperature are not directly interchangeable.
Geology / materials	Available geological or mineralogical indicators	Coverage and measurement methods may differ by body.

3.4 Processing pipeline

1. Inspect source metadata, licenses, spatial resolution, time coverage, units and coordinate reference systems.
2. Define a consistent region size and analysis scale for each comparison.
3. Convert compatible measures to shared units or standardized feature representations.
4. Derive terrain metrics from suitable elevation products; preserve provenance and processing parameters.
5. Flag missing or incomparable values, and preserve source identifiers and processing parameters in the feature tables.

4. Similarity Model and Evaluation

4.1 Baseline scoring approach

The minimum viable model should be transparent and easy to validate. For each candidate Earth region, calculate normalized differences from the selected planetary reference across the features that can be compared meaningfully. Assign documented weights to features according to the research objective.

$$D = \Sigma(w_i \times d_i) \qquad \text{Similarity} = 100 \times (1 - D)$$

Here, d_i is a feature-specific difference scaled to the interval [0, 1], and w_i is its importance weight. Weights must be non-negative and sum to 1, so D also lies from 0 to 1. The resulting score lies between 0 and 100. It describes similarity under the chosen features and assumptions; it is not a probability of suitability.

NORMALIZATION EXAMPLE

For a numeric feature, one possible bounded difference is $d_i = \min(1, |x_i - t_i| / R_i)$, where x_i is the Earth value, t_i is the target value and R_i is a documented comparison range. Other feature types may require different distance functions.

4.2 Recommended model-development sequence

Stage	Method	Purpose
Baseline	Weighted normalized distance	Clear, reproducible ranking for the MVP.
Sensitivity check	Vary weights within documented ranges	See whether rankings depend heavily on subjective weights.
Exploratory ML	Clustering or dimensionality reduction	Identify groups and visualize multi-feature patterns.
Advanced phase	Feature-specific uncertainty and robust ranking	Represent data quality and uncertainty more explicitly.

4.3 Handling missing or incomparable data

- Use only documented, comparable feature pairs when computing a given candidate score.
- Report the number of available features and a coverage indicator beside each score.
- Do not reward a candidate for having missing measurements; use minimum coverage rules or abstain from ranking it.
- Keep Earth-relative relief separate from planetary absolute elevation unless datums can be meaningfully transformed.
- Use appropriate illumination geometry; do not treat an Earth day/night cycle as equivalent to lunar or Martian lighting.

4.4 Validation plan

Check	Validation question
Data integrity	Are data sources, units, coordinate systems, scale and preprocessing steps documented?
Calculation tests	Do scores match hand-checked examples and boundary cases?
Ranking stability	Do rankings remain broadly stable under modest weight changes?
Coverage	What fraction of planned features is available for each candidate?
Interpretability	Can a user identify which features drive a score and which do not match?
Reproducibility	Can the same input and configuration reproduce the same output?

5. Data Sources and Data Readiness

The project should prioritize official data repositories and product documentation. The exact product should be selected only after checking its spatial coverage, resolution, units, calibration, licensing and suitability for the feature being compared.

Data family	Proposed source	Potential use / caution
Lunar topography	NASA LRO / LOLA [2]	Lunar elevation, slopes, roughness and related terrain products; inspect the specific product and resolution.
Mars images / terrain	NASA Planetary Data System; MRO / HiRISE products [3, 4]	Use appropriate imagery or derived digital terrain models; raw imagery is not automatically a DEM.
Earth observations	NASA Earthdata [5]	Discover suitable land-surface, climate and other Earth data. Check each product and spatial coverage.
Earth elevation	USGS 3D Elevation Program (3DEP) [6]	Useful terrain data for covered U.S. areas; global candidates may require other appropriate elevation sources.
Supporting geology / context	Relevant agency or peer-reviewed sources	Use documented geological or mineralogical products only when their coverage supports the comparison.

5.1 Data-readiness checklist

- Confirm each dataset can be accessed, used and processed within the hackathon time budget.
- Check coverage, resolution, units and whether the measurements represent comparable physical properties.
- Record source IDs, versions and transformations; preprocess a small documented subset for a reliable demo.

PRACTICAL IMPLEMENTATION CHOICE

Preprocess a small and documented subset of source data, then ship those feature tables with the demo. This improves reliability while keeping source links and provenance visible.

6. Technical Architecture and User Experience

6.1 Architecture

USER INTERFACE

Reference-site selection • feature controls • interactive map

ANALYSIS SERVICE

Feature preparation • normalization • ranking • explanation

EARTH DATA

Elevation • land surface • environment • geology when available

PLANETARY DATA

Lunar topography • Mars imagery / derived terrain • metadata

OUTPUTS

Ranked candidates • charts • source trail • coverage and uncertainty

6.2 Recommended technology stack

Layer	Technology	Reason
Processing	Python, Pandas, NumPy	Data wrangling and numerical analysis.
Raster / geospatial	Rasterio, GDAL, GeoPandas, Shapely	Work with rasters, vector regions and spatial operations.
Model	NumPy / scikit-learn	Transparent scoring first; exploratory ML later.
Interface	Streamlit	Fast dashboard development for a hackathon.
Maps and charts	Folium or Leaflet; Plotly	Interactive geographic exploration and comparison plots.
Collaboration	Git and GitHub	Version control, issue tracking and coordinated development.

6.3 Required screens

Screen	Minimum content
Mission dashboard	Selected reference, data coverage, current analysis status.
Earth explorer	Map of candidate regions with a clickable location profile.
Compare	Feature values side by side, with units and source/resolution notes.
Rankings	Ordered candidates, similarity score, feature contributions and coverage.
Evidence report	Methods, data links, assumptions, limitations and exportable summary.

7. Team Structure and Work Plan

7.1 Suggested five-person team

Role	Primary responsibilities	Deliverable
1. Team lead / science research	Translate challenge into scope; define reference regions; coordinate work and pitch.	Scope, research summary, presentation.
2. Data engineer	Find datasets; download, clean and document features; manage data files.	Reproducible data tables and provenance.
3. Geospatial analyst	Process rasters, derive terrain measures, define candidate Earth regions.	Feature extraction and map layers.
4. Data scientist	Implement scoring, missing-data handling, sensitivity checks and validation.	Ranking engine and test results.
5. Web developer / UI	Build dashboard, maps, charts and candidate details; integrate analysis.	Working demo and user workflow.

Roles are flexible. Cross-review the data pipeline and ranking calculations so that no single person is the only one who understands a critical component.

7.2 Suggested 48-hour schedule

Time window	Work package	Exit criterion
0-6 h	Research, define scope, shortlist datasets	Reference regions and feasible feature set agreed.
6-16 h	Data acquisition and preprocessing	Small validated feature table exists.
16-25 h	Similarity engine and tests	Rankings generated from real processed values.
25-36 h	Dashboard and visualization	Map, ranking and feature comparison work end to end.
36-43 h	Validation and integration	Source traceability, score checks and missing-data treatment reviewed.
43-48 h	Pitch, demo hardening and submission	Rehearsed demo, screenshots, repository and report ready.

TIME MANAGEMENT RULE

If data processing slips, reduce the number of sites or features. Do not replace unavailable data with invented values just to make the dashboard look complete.

7.3 Repository layout suggestion

```
terratwin/
  data/processed/  |  src/ingest/  |  src/features/  |  src/scoring/
  app/  |  tests/  |  docs/  |  README.md
```

8. Risks, Innovation and Success Criteria

8.1 Key risks and mitigations

Risk	Why it matters	Mitigation
Incompatible units / datums	Raw values may not mean the same thing across planetary bodies.	Compare relative measures or transform carefully; document all assumptions.
Resolution mismatch	Fine details visible in one dataset may not exist in another.	Use justified scales and disclose lost detail.
Missing features	A candidate could appear better because less data is available.	Report coverage and use minimum-data rules.
Arbitrary weights	Rankings could reflect personal preference rather than evidence.	Expose weights, document rationale and run sensitivity analysis.
Slow downloads	Large rasters could consume the entire event.	Preselect smaller regions and preprocess reproducibly.
Overclaiming outcomes	Similarity is not equivalent to safety or mission suitability.	Use calibrated language and require expert review for operational decisions.

8.2 What is innovative about TerraTwin?

The project should not claim that Earth analog research is new. Its potential contribution is the integration of a repeatable discovery workflow with configurable feature comparisons, transparent rankings, candidate-level explanations and explicit data-quality information.

- Repeatable discovery: apply the same documented process to more than one reference area.
- Explainable output: show which variables support or weaken each match.
- User-configurable objectives: investigate how priorities change the candidate ordering.
- Uncertainty awareness: make resolution, coverage and missing measurements visible.
- Open and reproducible workflow: preserve data provenance and document processing settings.

8.3 Proposed success criteria

Measure	MVP success condition
End-to-end function	At least one reference site can be compared against multiple Earth candidates using actual processed source data.
Traceability	Every displayed score links to inputs, features, weights and source products.
Score validation	Calculation tests pass on hand-checked cases and edge conditions.
Coverage transparency	Missing or incomparable variables are disclosed in the interface.
Usability	A new user can select a target and understand the ranking without reading source code.
Reproducibility	Another team member can regenerate the same results from the documented inputs and configuration.

8.4 Limitations

Earth cannot reproduce every lunar or Martian condition, including gravity, atmosphere, radiation, dust and thermal cycles. Rankings also depend on data, scale, features and weights; they describe specific analog characteristics, not a complete replica of another world.

9. Expected Deliverables and Future Roadmap

9.1 Hackathon deliverables

- Working TerraTwin web application with a reference-region selector.
- Processed feature tables derived from documented real datasets.
- An explainable similarity-ranking method with tested calculations.
- Interactive map and feature-comparison visualizations.
- Candidate profiles containing source links, coverage indicators and limitations.
- Project repository, setup instructions, technical diagram and short presentation.
- Demo script showing one lunar and one Martian comparison path, if data preparation allows.

9.2 Future roadmap

Phase	Potential enhancement
Phase 1: MVP	Two reference examples, a limited candidate set, transparent feature scoring and dashboard.
Phase 2: Broader coverage	More target regions, larger Earth screening areas, additional elevation sources.
Phase 3: More science	Better illumination modelling, geological/mineralogical features and uncertainty propagation.
Phase 4: Research workflow	Exportable reports, API access, saved configurations and reproducible analysis bundles.

9.3 Final recommendation

Build the smallest scientifically defensible product first. Prove that real source data can be processed into comparable features, that the ranking is reproducible, and that users can understand why a candidate was returned. Only then add more features, a larger study area or advanced machine-learning methods.

THE FIRST MILESTONE

Select one lunar reference region, one Martian reference region and three to five features that can be measured comparably. Process real data, generate a ranked candidate list and present the evidence behind each score.

10. Short Project Pitch

How can we prepare for exploring other worlds before we reach them? TerraTwin is a geospatial data science platform that uses open Earth, Moon and Mars datasets to identify and characterize terrestrial environments with similar measurable properties. It compares selected planetary reference regions with Earth candidates, ranks the candidates using transparent feature-based scoring, and visualizes both similarities and mismatches on an interactive map. By making the data, assumptions and uncertainty visible, TerraTwin aims to help researchers, students and exploration teams investigate where Earth can serve as a useful test environment for selected aspects of future lunar and Martian missions. It does not certify site safety or reproduce every condition of another world; it provides a reproducible starting point for further scientific investigation.

References and Official Resources

The following public resources provide challenge context and candidate data entry points. Accessed 13 October 2026. Product documentation should be consulted before analysis; a portal is not a guarantee that a specific dataset will fit the project requirements.

[1] NASA Space Apps Challenge 2026, challenge description reproduced in official team listings.
https://www.spaceappschallenge.org/2026/find-a-team/analogx/
Challenge statement on terrestrial analogs and open Earth, Moon and Mars data.

[2] NASA Science, Lunar Reconnaissance Orbiter - Lunar Orbiter Laser Altimeter (LOLA).
https://science.nasa.gov/mission/lro/lola/
Lunar topography, slopes, surface roughness and reflectance-related measurements.

[3] NASA Planetary Data System (PDS).
https://pds.nasa.gov/
Official archive entry point for planetary science data and product documentation.

[4] NASA Open Data Portal, MRO Mars High Resolution Imaging Science Experiment (HiRISE) EDR.
https://data.nasa.gov/dataset/mro-mars-high-resolution-imaging-science-experiment-edr-v1-0
Dataset description for Mars Reconnaissance Orbiter HiRISE image data; inspect product level before deriving terrain.

[5] NASA Earthdata.
https://www.earthdata.nasa.gov/
Entry point for NASA Earth-observation data discovery.

[6] U.S. Geological Survey, 3D Elevation Program (3DEP).
https://www.usgs.gov/3d-elevation-program/what-3dep
High-quality topographic and three-dimensional elevation data for covered U.S. areas.

Notes on evidence and status

This document is a proposed project report and implementation plan. It contains no newly computed location rankings, empirical accuracy results or verified site recommendations. Any such outputs should be added only after the project team downloads the cited data, documents preprocessing, validates the method and runs the analysis. The NASA Space Apps Challenge is a public event; TerraTwin is a student-project concept and is not an official NASA product or endorsement.